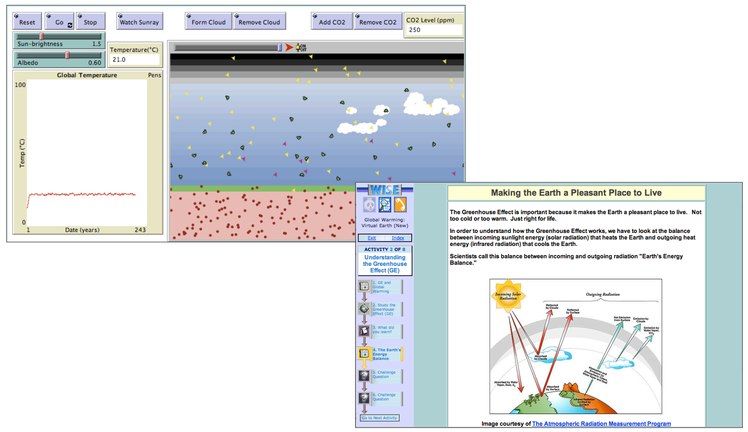
Evaluating Environmental Concepts in Student Drawing and Understanding Systematic Experimentation Behavior Using Technology

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RONALD E. McNAIR POST-BACCALAUREATE ACHIEVEMENT PROGRAM

Introduction

Web-Based Inquiry Science Environment (WISE) is a simple yet powerful online learning environment where students examine real world evidence and analyze current scientific controversies. This study examines student learning in a particular WISE project, *Climate Change: Virtual Earth*. The main goal of the project was to see if students who display more systematic behavior improve their post-test scores more than those who used less systematic behavior. Learning is measured by comparing the drawings that students generate in their pre-post assessments. Log data files showing how students manipulated variables in the simulation are coded for systematic patterns. This has the potential to reveal how systematic experimentation affects comprehension and the role of drawing in comprehension.



Methodology

Students were able to manipulate variables in a computer simulation called the "Greenhouse Model". The curriculum included online journal entries, videos, and challenge questions.

Research Question
Do students who display more systematic experimentation behavior show better comprehension than students with low systematicity from their pre- to post-test scores?

Participants
123 seventh grade students participated in Climate Change: Virtual Earth. The focus of the study was those students who included a drawing with their pre- and post-test, N=58.





Data Analysis

Written Responses

"Explain the Greenhouse Effect, making sure to use the terms solar energy, infrared energy, and greenhouse gases. You may also draw a picture."

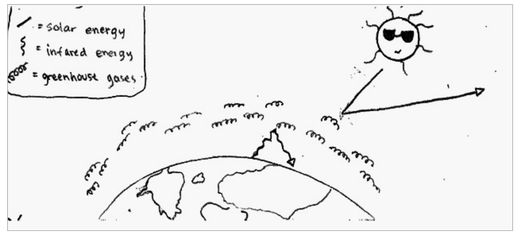
Example Pre-Test Responses:
Student 1: "It goes into the air and has bad gases that might cause bad things. Solar energy is very good because it uses energy from the sun not natural resources."

Student 2: "The greenhouse effect happens when solar energy and infrared energy combine with greenhouse gases and make the ozone layer weaker."

Example Post-Test Response:
Student 1: "I think that the greenhouse effect is good because the solar energy comes into the earth then it goes into the earth and becomes heat energy then it radiates and turns into infrared energy It makes the IR Energy go back into the earth."

Drawing Analysis: was divided into two parts; components and processes

Components: 3 possible points
1. Solar energy
2. Infrared energy
3. Greenhouse gases



Processes: 4 possible points
1. Solar energy enters atmosphere
2. Transformation to heat energy
3. Emission of infrared energy
4. Greenhouse gases "trap" infrared energy

Log Data Coding: was divided into three categories based on students experimentation behaviors

Systematic: students followed instructions for the simulation and only manipulated one variable at a time (controlling variables behavior – CVS)

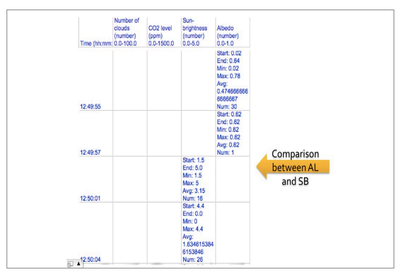
Unorganized: students control incorrect variable for 30 consecutive seconds or more

Exploratory: multiple variables are manipulated at the same time or within less than 30 seconds of each other

Systematic Experimentation



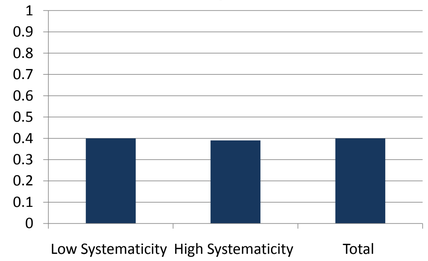
Exploratory Experimentation



Students were categorized by the number of systematic virtual experiments completed. Low = 1 or less ; High = 2 or more

Results

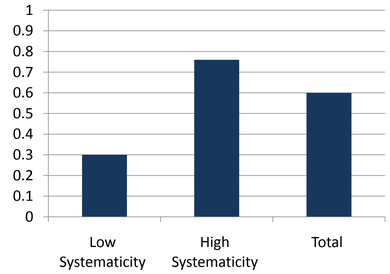
Written Response Gains



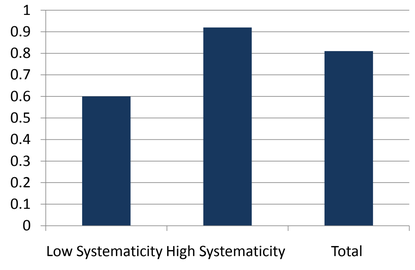
Analyses of students' written descriptions of the greenhouse effect, showed no significant differences between students who displayed high systematic behaviors versus those who were less systematic

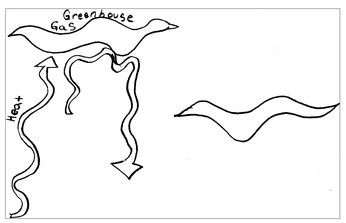
However, trends in the data show that students who were more systematic were able to create more complete drawings after they participated in the WISE project activities and conducted virtual experiments with the visualization.

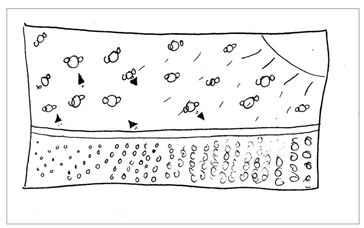
Drawing Processes Gain



Drawing Component Gains







Conclusions & Future Directions

Although the learning gains are not statistically significant further exploration should not be abandoned. The trends in the drawing data suggest that written explanation of systems or processes alone may not be the best way to measure students' understanding of complex scientific processes. Students may not be equipped with the vocabulary to describe the environmental phenomena but with assistance from teachers this can be improved. Further study should examine the learning that occurs during the unorganized runs within the data and on drawing as a strategy to express complex science processes. The partner dynamic is another area where further study would provide more insight into systematic experimentation behavior.

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